Neutrino and nucleus interaction

According to Fermi's theory, the neutron decomposes into a proton, an electron, and an electron neutrino. It's not certain that it is an antineutrino, as you can find on the web:
„As I said before though, it's not 100% certain that neutrinos aren't their anti-particles." Thus it's a Majorana fermion (Reference https://www.physicsforums.com/threads/neutron-proton-electron.5979/), which is a fermion that is its own antiparticle and obeys the Majorana equation. The form of the Majorana representation of gamma matrices:

$$i\Upsilon^{-\mu}\partial_\mu\psi - m\psi = 0 \quad (1)$$

Where all coefficients are real, we can take Ψ as a real function (for the Majorana fermions we must take Ψ as a real function). "They were hypothesized by Ettore Majorana in 1937. The term is sometimes used in opposition to a Dirac fermion, which describes fermions that are not their antiparticles. Except for the neutrino, all of the Standard Model fermions are known to behave as Dirac fermions at low energy (after electroweak symmetry breaking), and none are Majorana fermions. The nature of the neutrinos is not settled; they may be either Dirac or Majorana fermions".
http://inspirehep.net/record/15706/files/v14-n2-p69.pdf

We make the supposition that the ν anti-neutrino does not exist (its antiparticle).

In the reaction of Fermi's theory, we have neutron decay into a proton, electron, and an antineutrino; thus, a ν is a Majorana particle of energy 0.00038eV in the decay:

$$n + \tau \xrightarrow{decay} p + e + \nu + \mu \quad (2)$$

The reactions from the source nucleus and destination nucleus: An electronic neutrino ν collides with a proton, generating a neutron, a positron, and a μ neutrino, and evolves according to the steps below. At the nucleus, the reactions are continuous, and the transformations are as follows, where the abbreviations are: the sign. $\hat{\rceil}$ is emission and $\check{\rceil}$ is absorption, and the brackets are for grouping decay:

$$p + \lceil\nu \xrightarrow{transform} n + positron + \mu\hat{\rceil} \xrightarrow{decay} (p + e + \nu\hat{\rceil}) + positron = p + (e + positron) = p \quad (3)$$

And a τ neutrino will collide with a proton, generating the process of continuous migration of pairs of protons and neutrons, generating gravity, magnetron, and the μ neutrino that generates the hologram of space and A particle; We will analyze helium $_2He^4$ two pairs of binding spin proton-neutron. (4):

$$p + \lceil\tau + n \xrightarrow{transf} 2n + posit + \nu + mag + A\hat{\rceil} + \mu\hat{\rceil} \xrightarrow{decay} (p + e + \nu\hat{\rceil}) + n + posit = p + \mu\hat{\rceil} + e + posit + n = p + n$$

Thus, there must exist at least 2 nucleons in a specific amount of time of around $\sim 10^{-12}$ seconds to interact (mag) with two pairs of binding spin proton-neutron that change state with each other and exchange magnetron. As you can see, the initial proton in an atomic nucleus emits a ν neutrino, and a μ neutrino absorbs a ν neutrino and a τ neutrino to trigger the reaction, resulting in the pair of binding spin proton-neutron that change state with each other and exchange magnetron. In this way, the generation of gravity is provoked by the absorption and generation of a ν neutrino.

The gravity at the nuclear level between 2 nucleons is smaller than the electrostatic forces of repulsion between the 2 protons of Helium but is applied for a shorter period, about ten times less than the process of attraction between nucleons. The transformation of a neutron when receiving a τ neutrino:

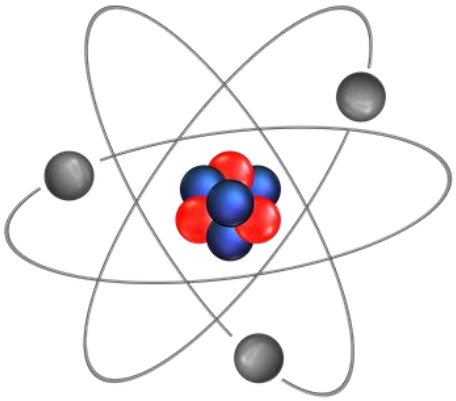
$n + \tau \xrightarrow{decay} p + e + \nu$ is in 4×10^{-12} s, and the transformation.

$p + \lceil\tau \xrightarrow{transform} n + positron + \mu\hat{\rceil}$ takes place in a $4 \cdot 10^{-12}$ sec while the attraction between nucleons is always present. Thus, the nucleons are constrained into a shell, and that explains the stability of the nucleus and the magic numbers of protons and neutrons. There's somehow a dance of nucleons into the nucleus that does create the gyromagnetic ratio of the atom's nuclear magnetic moment of the atom par-impair or impair-par. So the nucleons are most of the time neutrons, and for a short period, some of them turn into protons. The nucleons that change ν neutrino have quantum attraction forces that depend on a constant distance R = $3.9228 \cdot 10^{22}$ m and radius r:

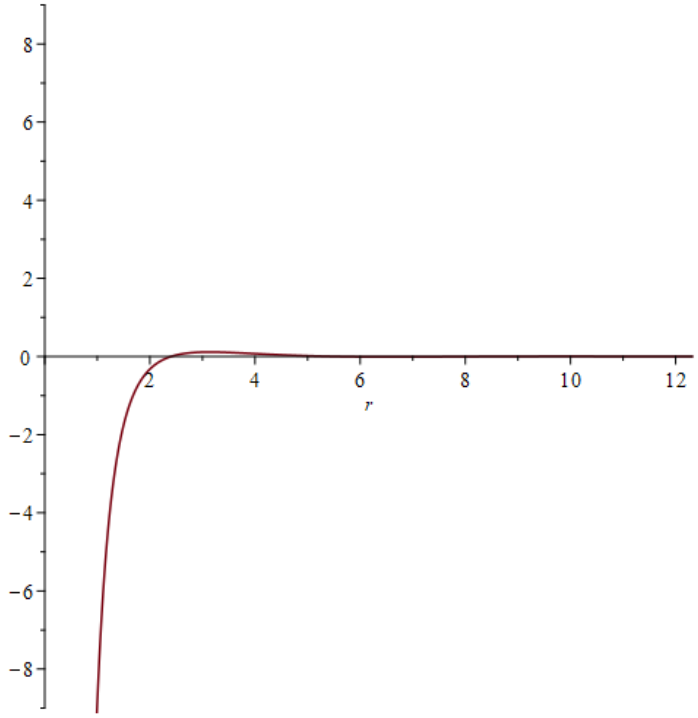
$$F_{g1} \sim \frac{3R^2}{14 \cdot \pi} \cdot \frac{1}{r^2} - \frac{R}{4 \cdot \pi} \cdot \frac{1}{r} + \frac{1}{14 \cdot \pi}$$

Thus, 1 Newton of force (102g) means the attraction between nucleons according to the file: grav.mw
The gluons carry the strong nuclear force. There are 5 fundamental forces: magnetic, electrostatic (Coulomb), gravity, the strong nuclear force, and probably the weak force(?). The strong nuclear force is responsible for the attraction between nucleons, but with the aid of magnetic spin force that aids the stability of the nucleus. The electrostatic repulsion between nucleons, combined with the gravitational force, all 4

balances the pulsating electrostatic force of repulsion that changes every ~ $4 \cdot 10^{-12}$ sec. The protons in the nucleus of Helium are ~ $4 \cdot 10^{-12}$ sec, repelled by the electrostatic forces and attracted by strong nuclear forces, magnetic and gravitational forces. For distances of ~10^{-15} m, the magnetic force of attraction for the particle that changes magnetrons is almost as strong as the electrostatic repulsion between protons, as you can see in the plot of the magnetron file Flux.pdf. It's known that the proton has μ_p 2.793 and the neutron has μ_n -1.913 for the nuclear magneton and electrostatic forces. Two nucleons that change magnetron are attracted with a force as in: https://michaelvio.github.io/Flux.pdf
Most of the time, nucleons are neutrons for ~ $4 \cdot 10^{-12}$ seconds, attracted by gravitational forces and magnetic forces, which, for the distance between nucleons r/2 < nucleon radius 0.55 +/- 0.09fm are significant.



The article: https://arxiv.org/pdf/2107.01441.pdf
In literature [2], we calculate the ratio of electrostatic force to gravitational force with the classical formula as $8.1 \cdot 10^{-37}$: Quantum Physics Vol IV- Berkley – Chapter 2-Magnitude of Physical Quantities (40a) page 73.
$\frac{F_g}{F_{es}} = \frac{G m_p{}^2}{4\pi\varepsilon e^2}$ where m is the proton mass, e is the charge of protons and G is Newton's gravitational constant.
$F_{es} = \frac{e^2}{4\pi\varepsilon r^2}$ For distance $r_0 = 2.4 \cdot 10^{-15}$ m, the repulsive electrostatic force is around ~40 N from the classical point of view, and for 4 fm, the distance between two protons in the Fe nucleus, we have 14 N. And the Magnetic Force (http://michaelvio.orgfree.com/Flux.pdf) at t = 0 is $Fm=H(r)/4\pi r^2$ plot for R=49 and r = 0.01..1:



We can see the Magnetic Force decrease very abruptly; thus, the magnetron is the particle of magnetic force that helps strong nuclear forces for the stability of the nucleus. The electrostatic repulsion between nucleons,

combined with strong nuclear forces and the attractive gravitational force and quantum magnetic forces, 4 balances the pulsating electrostatic force of repulsion that changes every ~ $4 \cdot 10^{-12}$ sec.
Thus, the quantum gravitational forces in Newtons between 2 nucleons that change ν neutrinos (gravitons) are: $F_{g1} \sim \frac{1.85 \cdot 10^{-64}}{r^2} \div \frac{4.09 \cdot 10^{-64}}{r^2}$
The weak nuclear force doesn't exist in the known form. The β-decay of the neutron is a consequence of the Standard Model that says that a Down Quark is turning into an Up Quark; it is a misunderstanding of the process. Quarks are the building blocks of matter, and they are very stable, not changing from one type to another. The Quark Information Theory explains that all stable particles are made of 3 quarks, and the information is not destroyed; it simply transforms.
Time is related to a τ neutrino and the sun generate τ neutrino, time, and gravity; τ neutrino is responsible for the time evolution process and at the Time Ionization State of the atom all process will freeze in the same as the piece of cheese in the kitchen will last a couple of days, in the fridge will last a couple of weeks, in the freezer will last a couple of months and into the Time Ionization State will last undefinably as long as the energy is supported for all the atoms into the peace of cease will remain in that state. We talk about organic matter and processes in the living world; a rock will last for a long time if there are no or limited erosive factors. In theory, it seems like about 3-5 wats/moll of substance for a beam of photons at the frequency ~313.5nm +/- 1%; you need to vary within this margin of error until the clock stops, either a mechanical or, better, a small electronic watch or even a fly, as you will see further in the message. The energy of the photons is E=hν and the Temporal Ionized Energy is E_T = 3.95eV…You may put a fly, *Drosophila melanogaster*, and paralyze it for a period longer than the life of the fly and recover it, so the life of the fly is doubled. Also, you can put a little mouse, a cat, or a dog there; the only difference is the total amount of energy…
The μ neutrino generated by atoms will provoke the duality of space and matter, which is a quantum of time wave and another quantum of time particle at the microscopic level.
The dual wave-particle behavior of the microscopic world is a consequence of quantum space… yes, space is alternately both wave and particle in consecutive quanta of time. Is a simple explanation of the fact that an electron entering through a double-slit experiment interfered with itself. Quantum Space gives reality and all particles the duality wave-particle which is generated in essence by quantum space particles. Space is one quantum of time is a wave, and one quantum of time is a particle with mass. That's why an electron passing through a double split interfered with him in the time quanta in which it's a wave and also in the double split if the conditions are right the width of the split is the order of size with a wavelength of the incident particle, so the interference process appears, and the particle has an associated wavelength (de Broglie) in the order of width split.
The Troitsk experiment at the link below shows that the mass of ν electronic is ~2 eV (-95%) https://www.researchgate.net/publication/51961903_An_upper_limit_on_electron_antineutrino_mass_from_Troitsk_experiment; thus the Gravitonic Ionization Energy should be 3.259eV and Time Ionization Energy ~3.95 eV.
As is said in the physics community, the proton can be transformed into a neutron only inside the nucleus; the transformation is circular, starting from a proton and regenerating the proton.
Several experiments probing whether the neutrino is a Majorana particle are currently underway.[1] [1] A. Franklin, *Are There Really Neutrinos?: An Evidential History* (Westview Press, 2004), p. 186
Quantum Physics Vol IV - Berkley – Chapter 2-Magnitude of Physical Quantities (40a) page 73 [2]

The mechanism of neutrino generation in matter and solid bodies is the absorption of excitation energy 2·Eν, 2·Eμ, or 2·Eτ in the nucleus and the generation of 2 neutrino particles 2·ν, 2·μ, or 2·τ. So, ionized another nucleus, and an avalanche phenomenon of generation occurs when two particles coherently excite the nucleus, and the nucleus generates 2 identical particles with the same energy. The neutrino, muon, and electron neutrino mechanisms are the same mechanism of action.
Graviton: 1) The 3.259eV energy of the photon generates 2 neutrinos of energy 1/2 => 1.6295eV, i.e., neutrinos with λ= 761nm, which in turn produce neutrinos with equal energy in the avalanche inside the solid body, generating antigravity. 2) Theoretically, for photons with λ= 761nm and energy 1.6295eV, after a burst of a few tens to hundreds of nm, it is gravitonically ionized with photons and neutrinos, but not in an avalanche; it can be experimented on sheets (foil) that will levitate. 3) In photons with λ= 253.6 nm and energy 4.8885eV, the generation of 3 neutrinos occurs, which produces levitation at higher light intensity,

but the phenomenon occurs very slowly; the probability of generating 3 neutrinos is lower, with a very precise frequency, but it is detectable; the body changes its weight.
Muon neutrino: 1) The 3.59eV energy of the photon generates 2 neutrinos of energy 1/2 => 1.795eV, i.e., neutrinos with λ= 690nm, which in turn produce neutrinos with equal energy in the avalanche inside the solid body, generating muon ionization. 2) Theoretically, photons with λ= 690 nm and energy 1.795 eV, after a few tens to hundreds of nm, it is muonically ionized with photons and neutrinos, but not in an avalanche; it can be experimented on sheets (foil) that will enter ionization. 3) With photons with λ= 230 nm and energy 5.385 eV, the generation of 3 neutrinos occurs, which produces ionization at higher light intensity, but the phenomenon occurs very slowly; the probability of generating 3 neutrinos is lower, with a very precise frequency, but it is detectable; the body changes its state.
Tau neutrino: 1) The 3.95eV energy of the photon generates 2 neutrinos of energy 1/2 => 1.875eV, i.e., neutrinos with λ= 627nm, which in turn produce neutrinos with equal energy in the avalanche inside the solid body, generating temporal ionization. 2) Theoretically, for photons with λ= 627nm and energy 1.975 eV, for a few tens to hundreds of nm it is temporally ionized with photons and neutrinos but not in an avalanche; it can be experimented on sheets (foil) that will enter ionization. 3) With photons with λ= 209 nm and energy 5.925eV, the generation of 3 neutrinos occurs, which produces ionization at higher light intensity, but the phenomenon occurs very slowly; the probability of generating 3 neutrinos is lower, with a very precise frequency, but it is detectable; the body changes its state.

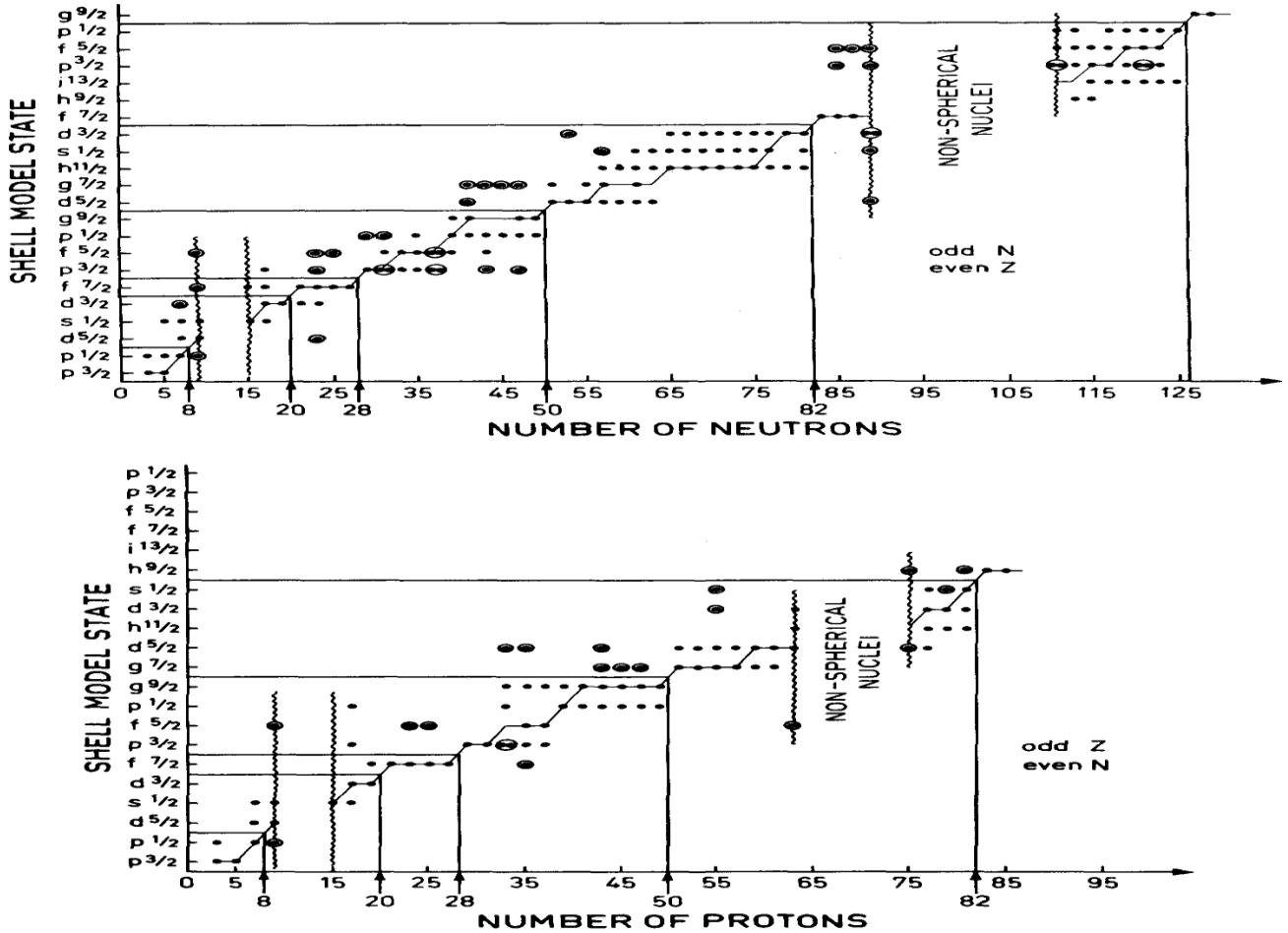


Figure 2-24 *One-particle interpretation of low-energy levels in odd-A nuclei. The data for the figure are taken from Nuclear Data Sheets. The regions of stable ellipsoidal deformations are excluded, as indicated by the wavy lines. We wish to thank G. T. Ewan for help in the preparation of the figure.*

31. Aug. 2026